Keeping Experts Honest*

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Abstract

Peer review and grant allocation routinely involve reviewers whose conflicts of interest are observable to the decision-maker: extensive self-citation, recent co-authorship, methodological commitments to a particular research program. Current institutional responses to such conflicts are coarse. The reviewer is either fully included, with the editor discounting her input informally, or fully recused. This paper studies an intermediate option. In an infinite-horizon, discrete-time model with noisy binary signals and a state-independent bias known to the principal, the optimal mechanism for using a conflicted expert is a randomized grim-trigger. The principal rubber-stamps the expert's recommendation by default, and after any funded project that turns out to be bad, commits with a calibrated probability to permanently ignore the expert from then on. The probability has a closed form in the signal precision, prior, bias, and discount factor; the mechanism is optimal within the full class of history-dependent mechanisms under a natural patience condition. The result formalizes a missing option in conflict-of-interest policy: probabilistic use of a conflicted reviewer, calibrated to the severity of the conflict.

Keywords: repeated games, mechanism design without transfers, expert advice, cheap talk, grim-trigger strategies, peer review.

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1 Introduction

Conflicts of interest in peer review come in many forms. A reviewer is asked to evaluate a paper that cites her work extensively. An experimentalist is asked to evaluate a study testing her own theoretical prediction. A methodologist is asked to evaluate a paper using a framework she has spent her career critiquing. In each case, the editor or program officer can identify the conflict in advance, but the institutional menu of responses is short. Severe conflicts trigger recusal: the reviewer is replaced with someone less informed but more neutral. Mild conflicts are discounted informally. Anything in between gets handled case-by-case, with no real principle behind the discounting.

Li (2017) quantifies the underlying trade-off at the NIH. Reviewers whose expertise is closer to a proposal are both more informed and more biased about it. Expertise wins on net, but bias is large enough to matter. The institutional implication is clear: in a wide range of cases, neither full inclusion nor full recusal is the right answer. The reviewer should be used, but used carefully. What does "carefully" mean when monetary transfers are unavailable and the only currency the principal has is the right to decide whether to listen?

I take up this question in a discrete-time, infinite-horizon model. Each period, a binary-quality project arrives, and an expert observes a noisy binary signal about it before recommending an action. The expert's preferences put weight $\lambda \in [0, 1]$ on the principal's payoff and $1-\lambda$ on whether the project gets funded, with λ common knowledge. Both players share a discount factor $\delta \in (0, 1)$. The principal commits at $t = 0$ to a mechanism mapping the public history of recommendations and realized qualities into current and future funding decisions.

The optimal mechanism turns out to be simple. Call it the *Almost-Unforgiving Information Transmission* mechanism, or AUNT. The principal follows the expert's recommendation as long as she remains "active." After any funded project that turns out to be bad, the principal moves to an absorbing "silent" state with probability $\varepsilon(\lambda)$ and stays active otherwise. In the silent state, no project is ever funded again, regardless of recommendation. The triggering probability $\varepsilon(\lambda)$ is chosen to make the expert exactly indifferent between truth-telling and lying after a bad signal, and it has a closed-form expression in the four primitives of the model.

AUNT is the missing third option in conflict-of-interest policy. Pure rubber-stamping corresponds to $\varepsilon = 0$ and is optimal only when the reviewer's preferences are already aligned enough with the principal's that no discipline is needed. Pure recusal corresponds to $\varepsilon = 1$, the limit case where any funded mistake is permanent. The calibrated AUNT mechanism sits between them. Severe conflicts get harsh triggers; mild conflicts get gentle ones. The threshold $\bar{\lambda}(p, q)$ at which ε falls to zero marks the boundary above which the conflict is no longer binding. The randomization itself is essential. The expert is included on the equilibrium path with probability one, but the small chance of permanent exclusion after any mistake is what sustains truthful reporting.

Three features of AUNT are worth flagging. It is stationary, so the same rule applies every period. Its history dependence is minimal, with the relevant state binary: active or silent. It conditions only on the realized qualities of *funded* projects, never on rejected ones, which matters in applications like grant review where rejected proposals produce no observable outcome. The trigger

is harsh but rare. Along the equilibrium path it fires only after honest but unlucky recommendations, and even then only with probability ε. For modest bias and reasonable signal precision, an honest expert can expect to remain active for a long time.

The optimality argument has two parts. A revelation-principle lemma restricts attention to mechanisms in which the expert reports truthfully each period. Among truthful mechanisms, the principal's payoff is proportional to the rate at which he follows the expert, so he wants this rate as high as the incentive constraints allow. Because the expert's bias is one-sided, with temptation to lie only after a bad signal, the binding constraint is unique. The bang-bang logic of Abreu et al. (1990) then forces optimal continuation values to the corners of the feasible payoff set. Either the expert is left in the active state or she is sent to silence. AUNT randomizes between these corners at the rate that just binds the relevant truth-telling constraint.

The paper proceeds as follows. Section 2 discusses related work. Section 3 sets up the model. Section 4 characterizes the first-best benchmark and shows that no mechanism attains it under imperfect signals. Section 5 defines AUNT, derives $\varepsilon(\lambda)$, and proves optimality among all history-dependent mechanisms. Section 6 concludes. An appendix collects the near-first-best result for almost-perfect precision and proofs deferred from the main text.

2 Related Literature

The paper contributes to three connected strands: repeated cheap talk and expert advice, dynamic incentive design without monetary transfers, and review strategies in repeated games with imperfect monitoring. It also speaks to a small applied literature on scientific grant allocation.

Cheap talk and dynamic communication. The point of departure is Crawford and Sobel (1982), in which biased communication yields only coarse partitional equilibria. Sobel (1985) showed that with repeated interaction a single expert can build and then strategically exploit credibility; the AUNT mechanism is the principal's optimal commitment response to exactly this dynamic. Bénabou and Laroque (1992) added noisy private information to a reputational framework, showing that signal imperfection lets manipulation persist—the same imperfection that motivates this paper's probabilistic punishment. Lipnowski and Ravid (2020) characterize sender-preferred equilibria with state-independent ("transparent") motives, providing the static benchmark this dynamic mechanism improves upon. Margaria and Smolin (2018) establish a folk theorem for dynamic cheap talk with state-independent sender payoffs; the present paper sharpens that existence result to an exact optimal-mechanism characterization. Closest in environment, Kivinen and Kuzmics (2025) ask which grim-trigger-like equilibria survive renegotiation in the same setting; their negative result and the positive characterization here are complementary—under commitment a grim-trigger mechanism is receiver-optimal, while under renegotiation-proofness it is not.

Dynamic incentives without monetary transfers. When transfers are unavailable, the principal's instruments are current and future allocations or decisions. Deb et al. (2018) characterize an

optimal mechanism for evaluating a forecaster of unknown precision; the friction here is bias rather than precision, and the expert's type is known. Guo and Hörner (2020) study dynamic allocation without money, finding optimal mechanisms in which inefficiency is backloaded and the agent is eventually terminated—a structural parallel to AUNT's absorbing punishment.

Review strategies, reputation, and APS. The proof uses self-generation and bang-bang continuation values in the tradition of Abreu et al. (1990). AUNT is a review strategy in the spirit of Radner (1985), with a randomized trigger reminiscent of Green and Porter (1984)'s oligopoly construction. The "simple-is-optimal" pattern echoes Fuchs (2007), who shows that simple "fire" mechanisms dominate elaborate ones in repeated moral hazard with private evaluations. An alternative discipline tool is reputation rather than explicit commitment: Ely and Välimäki (2003) show that reputational concerns can destroy surplus, and Cripps et al. (2004) prove that reputation effects are impermanent under imperfect monitoring. These results motivate the commitment-based approach taken here: the principal cannot rely on endogenous reputation alone.

Science funding. The motivating application connects to a small theory literature on grant allocation; Carnehl et al. (2025) provide a comprehensive overview of mechanism design for scientific grants. On the empirical side, Li (2017) provides the most direct evidence for the expertise-versus-bias trade-off modeled here.

3 Model

3.1 Environment

Time is discrete and the horizon is infinite: $t = 0, 1, 2, \ldots$. There is one long-lived *principal* and one long-lived *expert* (equivalently, *agent* or *advisor*). Both discount the future at the common rate $\delta \in (0, 1)$.

Each period t, a project of quality $q_t \in \{0, 1\}$ arrives. Qualities are independent and identically distributed across periods with $\mathbb{P}(q_t = 1) = q \in (0, 1)$. Before the principal decides whether to fund the project, the expert privately observes a signal $\eta_t \in \{0, 1\}$ satisfying

$$\mathbb{P}(\eta_t = q_t \mid q_t) = p$$

for a precision parameter $p \in (1/2, 1]$. The signal is conditionally independent across periods given the sequence of qualities. The precision p is common knowledge and does not depend on the expert's effort.

After observing η_t, the expert submits a recommendation $r_t \in \{0, 1\}$, where $r_t = 1$ is interpreted as a recommendation to fund. A public randomization device x_t drawn uniformly from $[0, 1]$ realizes simultaneously. The principal then makes a funding decision $d_t \in \{0, 1\}$. At the end of the period

the quality q_t is publicly revealed.[1]

The *public history* at the start of period t is

$$h^t \equiv \big((r_0, q_0, x_0), (r_1, q_1, x_1), \ldots, (r_{t-1}, q_{t-1}, x_{t-1})\big),$$

with $h^0 = \varnothing$. Let $\mathcal{H}$ denote the set of all finite public histories. A *mechanism* is a collection of functions

$$d = \big\{\rho^r(h^t)\big\}_{h^t \in \mathcal{H},\ r \in \{0,1\},\ t \geq 0},$$

where $\rho^r(h^t) \in [0,1]$ is the probability that the principal funds the project at period t after history h^t and recommendation r. Implicitly, the public randomization draw x_t is used to implement any required randomization over d_t. The principal commits to d at $t = 0$.

An *expert strategy* is a collection of functions $\sigma_t(h^t, \eta_t; d) \in [0,1]$ giving the probability of reporting $r_t = 1$ conditional on the public history h^t, the signal realization η_t, and the mechanism d. Because signals are i.i.d. and the mechanism is public, it is without loss of generality to condition only on the public history (and the current signal) rather than on the expert's private history of past signals.

3.2 Payoffs

The principal's period payoff is

$$u^P(d_t, q_t) = \begin{cases} 1 & \text{if } d_t = 1 \text{ and } q_t = 1, \\ -1 & \text{if } d_t = 1 \text{ and } q_t = 0, \\ 0 & \text{if } d_t = 0, \end{cases}$$

and her objective is to maximize

$$U^P(d) = (1-\delta)\,\mathbb{E}\left[\sum_{t=0}^{\infty} \delta^t\, u^P(d_t, q_t)\right].$$

The expert's period payoff is

$$u(d_t, q_t) = \lambda\, u^P(d_t, q_t) + (1-\lambda)\,\mathbf{1}\{d_t = 1\},$$

where $\lambda \in [0,1]$ is the *alignment parameter*, common knowledge. When $\lambda = 1$ the expert's preferences coincide with the principal's; when $\lambda = 0$ the expert values only whether projects are funded, regardless of quality. The expert's continuation payoff at history h^t with current signal η_t under

[1] For the optimal mechanism below, only the quality of *funded* projects needs to be observed. See Remark 5.

mechanism d and strategy σ is

$$U_t(h^t, \eta_t; d, \sigma) = (1-\delta)\,\mathbb{E}\left[\sum_{s\geq t} \delta^{s-t}\, u(d_s, q_s) \,\middle|\, h^t, \eta_t\right].$$

Bias is one-sided: the expert is weakly biased toward funding. A symmetric treatment of the opposite bias is immediate and formalized in Proposition 1.

I adopt the tie-breaking convention that an indifferent expert reports truthfully: if U_t from $r_t = \eta_t$ equals U_t from $r_t \neq \eta_t$, the expert chooses $r_t = \eta_t$. This convention is inessential for the results.

3.3 Solution Concept and Maintained Assumption

The solution concept is *perfect Bayesian equilibrium* (PBE): the principal chooses a mechanism d maximizing $U^P(d)$ anticipating the expert's optimal response, and at every history and signal realization the expert maximizes her continuation payoff given the mechanism.

Throughout the analysis I assume that signal precision and prior are such that the expert's information is genuinely useful for the principal's decision:

Assumption 1 (Informative signals)**.** $1 - p < q < p$.

If $q \leq 1-p$ the principal would optimally reject every project regardless of signal; if $q \geq p$ he would accept every project. In both edge cases the signal is payoff-irrelevant and mechanism design is trivial.

For convenience, define

$$\kappa \equiv \mathbb{P}(\eta_t = 1) = pq + (1-p)(1-q),$$

and note that $\kappa \in (0, 1)$.

4 First-Best and Impossibility

I begin by establishing the principal's first-best benchmark and a sharp impossibility result for imperfect precision.

Definition 1 (First-best)**.** A mechanism d achieves the principal's *first-best* if $U^P(d)$ equals the principal's expected payoff from a counterfactual in which he directly observes the signal realizations $(\eta_t)_{t\geq 0}$.

Under Assumption 1, the first-best policy funds the project if and only if $\eta_t = 1$, yielding per-period expected payoff

$$U^{FB} = \mathbb{P}(\eta_t = 1, q_t = 1) - \mathbb{P}(\eta_t = 1, q_t = 0) = pq - (1-p)(1-q) = p + q - 1 > 0.$$

Consider the simple mechanism that implements whatever the expert recommends:

Definition 2 (Rubber-stamping)**.** The *rubber-stamping mechanism* d^{RS} is defined by $\rho^r(h^t) = r$ for all $r \in \{0, 1\}$ and all $h^t \in \mathcal{H}$.

Under d^{RS}, the expert's signal $\eta_t = 1$ always induces truthful reporting, since reporting $r_t = 1$ weakly dominates $r_t = 0$ stage-by-stage (the continuation value is unchanged by the report). After $\eta_t = 0$, the expert's expected stage payoff from lying ($r_t = 1$) is

$$\frac{\mathbb{P}(\eta_t = 0, q_t = 1)}{\mathbb{P}(\eta_t = 0)} \cdot 1 + \frac{\mathbb{P}(\eta_t = 0, q_t = 0)}{\mathbb{P}(\eta_t = 0)} \cdot (1 - 2\lambda) = \frac{(1-p)q + (1-2\lambda)p(1-q)}{1-\kappa},$$

which exceeds the zero stage payoff from truthful reporting whenever

$$\lambda < \bar{\lambda}(p, q) \equiv \frac{(1-p)q + p(1-q)}{2p(1-q)} = \frac{1-\kappa}{2p(1-q)}.$$

Thus rubber-stamping achieves the first-best if and only if $\lambda \geq \bar{\lambda}(p, q)$: when alignment is high enough, the expert's stage preferences already align with truthful reporting. The following theorem shows that if rubber-stamping fails, no mechanism can attain the first-best unless the signal is perfect.

Theorem 1 (Impossibility of first-best)**.** *Suppose $p < 1$ and $\lambda < \bar{\lambda}(p, q)$. Then no mechanism achieves the principal's first-best.*

Proof. The first-best policy funds if and only if $\eta_t = 1$, which is observationally indistinguishable from funding if and only if $r_t = 1$ when $r_t = \eta_t$ is played. Any mechanism that attains the first-best therefore induces (with probability one on the equilibrium path) funding decisions satisfying $d_t = \eta_t$. Under truthful reporting this requires $d_t = r_t$, i.e. rubber-stamping on the equilibrium path. But under rubber-stamping with $\lambda < \bar{\lambda}(p, q)$, truthful reporting fails after $\eta_t = 0$ as argued above, a contradiction. Any non-rubber-stamping implementation either fails to follow some $\eta_t = 1$ signal (losing an expected benefit of $\mathbb{P}(\eta_t = 1, q_t = 1) - \mathbb{P}(\eta_t = 1, q_t = 0) > 0$) or funds some $\eta_t = 0$ signal (losing an expected $\mathbb{P}(\eta_t = 0, q_t = 0) - \mathbb{P}(\eta_t = 0, q_t = 1) > 0$), and therefore also fails to attain the first-best. □

Remark 1 (Perfect precision)**.** When $p = 1$, the *Unforgiving Information Transmission mechanism* d^{UNIT}, defined below by setting $\varepsilon = 1$ in the AUNT mechanism, achieves the first-best for any λ provided $\delta \geq 1/(1+q)$. Appendix A extends this observation to a continuity result at $p = 1$.

5 The AUNT Mechanism and Its Optimality

In light of Theorem 1, the focus is on second-best mechanisms when $\lambda < \bar{\lambda}(p, q)$ and $p < 1$.

5.1 Definition and Value

Definition 3 (AUNT mechanism)**.** Fix $\varepsilon \in [0,1]$. The *Almost-Unforgiving Information Transmission mechanism with parameter* ε, denoted $d^{\text{AUNT}}(\varepsilon)$, is defined as follows. Starting from a state labeled "active," at each period:

(i) If the current state is *active*, the principal sets $d_t = r_t$ (rubber-stamping). If $r_t = 1$ and $q_t = 0$, the state transitions to *absorbing* with probability ε, using the public randomization device x_t; otherwise the state remains active.

(ii) If the current state is *absorbing*, $d_t = 0$ for all future periods.

Two special cases are instructive. With $\varepsilon = 0$, AUNT reduces to rubber-stamping. With $\varepsilon = 1$, AUNT reduces to UNIT: the principal permanently ignores the expert after any incorrect positive recommendation.

Under $d^{\text{AUNT}}(\varepsilon)$, truthful reporting generates the following continuation value for the expert in the active state. Let σ_R^* denote this value. By stationarity (the mechanism is time-invariant within the active state and qualities are i.i.d.),

$$\sigma_R^* = (1-\delta)\big[pq + (1-2\lambda)(1-p)(1-q)\big] + \delta\big[1 - \varepsilon(1-p)(1-q)\big]\sigma_R^*,$$

since under truthful play the stage payoff comes only from funded projects (which occur when $\eta_t = 1$, yielding the bracket), and punishment occurs exactly when $\eta_t = 1$ while $q_t = 0$, which happens with probability $(1-p)(1-q)$. Solving,

$$\sigma_R^* = \frac{(1-\delta)\big[pq + (1-2\lambda)(1-p)(1-q)\big]}{(1-\delta) + \delta\varepsilon(1-p)(1-q)}. \tag{1}$$

5.2 The Optimal Punishment Probability

The principal chooses ε to make the expert willing to report truthfully at the binding signal realization. When $\lambda < \bar{\lambda}(p,q)$, the expert is tempted to report $r_t = 1$ after $\eta_t = 0$; the constraint $r_t = \eta_t = 1$ is slack (see Lemma 2 below). Comparing the expected payoffs from $r_t = 0$ versus $r_t = 1$ conditional on $\eta_t = 0$:

$$\begin{aligned}
&\text{Truthful } (r_t = 0): \quad \delta\sigma_R^*, \\
&\text{Deviate } (r_t = 1): \quad (1-\delta)\cdot\frac{(1-p)q + (1-2\lambda)p(1-q)}{1-\kappa} \\
&\qquad\qquad + \delta\left[\frac{(1-p)q}{1-\kappa} + \frac{p(1-q)}{1-\kappa}(1-\varepsilon)\right]\sigma_R^*.
\end{aligned}$$

Simplifying, the incentive constraint Truthful $\geq$ Deviate becomes

$$\delta \cdot \varepsilon \cdot p(1-q) \cdot \sigma_R^* \geq (1-\delta)\big[(1-p)q + (1-2\lambda)p(1-q)\big]. \tag{2}$$

Substituting (1) and simplifying using the algebraic identity

$$p \cdot \big[pq + (1-2\lambda)(1-p)(1-q)\big] - (1-p) \cdot \big[(1-p)q + (1-2\lambda)p(1-q)\big] = q(2p-1)$$

(which follows by direct computation), constraint (2) reduces to

$$\varepsilon \geq \varepsilon(\lambda) \equiv \frac{(1-\delta)\big[(1-p)q + (1-2\lambda)p(1-q)\big]}{\delta\, q(1-q)(2p-1)}. \tag{3}$$

The principal prefers smaller ε (less distortion on-path), so he sets $\varepsilon = \varepsilon(\lambda)$ at equality. Note $\varepsilon(\lambda)$ is strictly decreasing in λ and satisfies $\varepsilon(\bar{\lambda}(p,q)) = 0$, consistent with rubber-stamping being first-best-optimal at $\lambda = \bar{\lambda}$.

Lemma 1 (Feasibility of $\varepsilon(\lambda)$)**.** *$\varepsilon(\lambda) \leq 1$ for all $\lambda \in [0, \bar{\lambda}(p,q)]$ if and only if*

$$\delta \geq \delta^*(p,q) \equiv \frac{1-\kappa}{p(1-q) + q(1-\kappa)}. \tag{4}$$

Proof. Since $\varepsilon(\lambda)$ is decreasing in λ, $\varepsilon(\lambda) \leq 1$ for all $\lambda \in [0, \bar{\lambda}(p,q)]$ if and only if $\varepsilon(0) \leq 1$. Setting $\lambda = 0$ in (3):

$$\varepsilon(0) = \frac{(1-\delta)(1-\kappa)}{\delta\, q(1-q)(2p-1)}.$$

The inequality $\varepsilon(0) \leq 1$ rearranges to $(1-\delta)(1-\kappa) \leq \delta\, q(1-q)(2p-1)$, i.e.

$$\delta \geq \frac{1-\kappa}{(1-\kappa) + q(1-q)(2p-1)}.$$

Direct computation (expanding $1-\kappa = p(1-q) + q(1-p)$) gives $(1-\kappa) + q(1-q)(2p-1) = p(1-q) + q(1-\kappa)$, yielding (4). □

Remark 2 (Impatient principal)**.** When $\delta < \delta^*(p,q)$, the trigger probability $\varepsilon(\lambda)$ required by the binding (TT0) constraint exceeds 1 and AUNT is infeasible. The principal's best feasible response in that regime is UNIT ($\varepsilon = 1$), under which (TT0) holds with strict slack at every λ for which UNIT is itself incentive-compatible. The resulting principal payoff is strictly below the AUNT value when $\delta \geq \delta^*$, reflecting the fact that insufficient patience prevents the principal from calibrating punishment to exactly bind the incentive constraint. Theorem 3 characterizes the limit of this regime as $p \to 1$.

Remark 3. At $p = 1$, $\kappa = q$ and (4) simplifies to $\delta \geq 1/(1+q)$, recovering the almost-perfect-precision condition of Appendix A. For $p < 1$, the patience threshold is strictly higher: as the signal becomes noisier, the on-path cost of punishment grows relative to its deterrent value, and greater patience is needed to sustain the mechanism.

Under Assumption 1 and the patience condition (4), the principal's payoff from $d^{\text{AUNT}}(\varepsilon(\lambda))$ is

$$U^P\big(d^{\text{AUNT}}(\varepsilon(\lambda))\big) = \frac{(1-\delta)(p+q-1)}{(1-\delta)+\delta\,\varepsilon(\lambda)(1-p)(1-q)}, \tag{5}$$

since per-period the principal earns the first-best flow $p+q-1$ until an absorbing punishment occurs, which happens each period (conditional on still being active) with probability $\varepsilon(\lambda)(1-p)(1-q)$. Figure 1 illustrates how $\varepsilon(\lambda)$ varies with the alignment, precision, and patience parameters.

The other remaining incentive check is that the expert does not prefer to misreport after $\eta_t = 1$.

Lemma 2 (Truth-telling at $\eta_t = 1$)**.** *Under $d^{AUNT}(\varepsilon)$ with $\varepsilon \le 1$ and the promise-keeping continuation σ_R^* in (1), the expert weakly prefers $r_t = 1$ to $r_t = 0$ when $\eta_t = 1$.*

Proof. The constraint is:

$$(1-\delta)\cdot\frac{pq+(1-2\lambda)(1-p)(1-q)}{\kappa} + \delta\left[\frac{pq}{\kappa} + \frac{(1-p)(1-q)(1-\varepsilon)}{\kappa}\right]\sigma_R^* \ge (1-\delta)\cdot 0 + \delta\,\sigma_R^*.$$

Using $pq + (1-p)(1-q) = \kappa$, the coefficient of σ_R^* on the left simplifies to $1-(1-p)(1-q)\varepsilon/\kappa$. Subtracting $\delta\sigma_R^*$ from both sides and multiplying by κ:

$$(1-\delta)\big[pq+(1-2\lambda)(1-p)(1-q)\big] \ge \delta\,\sigma_R^*(1-p)(1-q)\,\varepsilon.$$

Substituting (1) and canceling the common numerator factor (which is positive when $\lambda < 1/2 + pq/[2(1-p)(1-q)]$, a condition implied by $\lambda < \bar{\lambda}(p,q) < 1$):

$$(1-\delta) + \delta\,\varepsilon(1-p)(1-q) \ge \delta\,\varepsilon(1-p)(1-q),$$

which always holds. □

5.3 Optimality within the Class of All Mechanisms

The main result establishes that AUNT with parameter $\varepsilon(\lambda)$ is optimal within the full class of feasible mechanisms.

Theorem 2 (Optimality of AUNT)**.** *Fix $p \in (1/2, 1)$, $q \in (1-p, p)$, $\lambda \in [0, \bar{\lambda}(p,q))$, and $\delta \ge \delta^*(p,q)$. The mechanism $d^{AUNT}(\varepsilon(\lambda))$ maximizes $U^P(d)$ over all feasible mechanisms.*

The proof proceeds in four steps. First, a revelation-principle-style lemma restricts attention to truthful mechanisms. Second, it is shown that after an on-path truthful report of $r_t = 0$, the continuation values after $q_t = 1$ and $q_t = 0$ can without loss be equalized. Third, the binding incentive constraint must be the one at $\eta_t = 0$. Finally, given these reductions, AUNT is the unique optimum up to payoff-irrelevant modifications.

I introduce notation for continuation values. For a truthful mechanism d and public history h^t, let $V(h^t) \in \mathbb{R}$ denote the expert's continuation payoff from period t forward under truthful play. For $r \in \{0,1\}$ and $q \in \{0,1\}$, let $V_q^r(h^t)$ denote the expert's continuation from period $t+1$ forward after history h^t, report $r_t = r$, and realized quality $q_t = q$ (along the equilibrium path of truth-telling). I use the shorthand

$$\sigma_R^r(h^t) = V_r^r(h^t), \qquad \sigma_P^r(h^t) = V_{1-r}^r(h^t),$$

so that "R" denotes the continuation when the recommendation matches the realized state ("Reward"), and "P" denotes the continuation when the recommendation mismatches ("Punishment"). Note that when $r = 0$ the principal does not fund, so "mismatch" means the expert mistakenly recommended rejection of a good project.

Lemma 3 (Revelation)**.** *There exists an optimal mechanism d^* under which the expert truthfully reports $r_t = \eta_t$ at every history on the equilibrium path.*

Proof. Let $(\tilde{d}, \tilde{\sigma})$ be any PBE with principal payoff U^P, where $\tilde{\sigma}(h^t, \eta) \in \Delta(\{0,1\})$ denotes the expert's (possibly mixed) report distribution at history h^t and signal η.

Define the direct mechanism d^* by

$$\rho^{*r}(h^t)(d) \;\equiv\; \sum_{r' \in \{0,1\}} \tilde{\sigma}(h^t, r)(r')\, \tilde{\rho}^{r'}(h^t)(d), \qquad r \in \{0,1\},$$

and extend continuation mechanisms recursively by the same composition. Intuitively, d^* responds to reported signal η exactly as $\tilde{d}$ responds to the expert's $\tilde{\sigma}$-distributed report when the expert observes signal η.

Consider σ^* given by truthful reporting, $\sigma^*(h^t, \eta) = \eta$. By construction, the joint distribution of (η_t, r_t, d_t, q_t) together with the public history under (d^*, σ^*) at every t coincides with that under $(\tilde{d}, \tilde{\sigma})$; iterating, the discounted payoff streams of both players coincide. In particular, the principal's payoff under (d^*, σ^*) equals U^P.

To verify that σ^* is a best response under d^*, fix any history h^t, signal η, and alternative report η'. Reporting η' in d^* generates, at every continuation history, the same joint distribution over $(d_t, q_t, \text{future continuation})$ as playing the action distribution $\tilde{\sigma}(h^t, \eta')$ in $\tilde{d}$. The expert's continuation utility from reporting η' in d^* at signal η therefore equals her continuation utility in $\tilde{d}$ at signal η if she were to play $\tilde{\sigma}(h^t, \eta')$ instead of $\tilde{\sigma}(h^t, \eta)$. Since $\tilde{\sigma}$ is part of a PBE, $\tilde{\sigma}(h^t, \eta)$ is a best response at signal η, so this continuation utility is weakly smaller than the one from reporting η truthfully. Hence truth-telling is a best response at every (h^t, η); the tie-breaking convention selects it whenever the expert is indifferent.

Thus (d^*, σ^*) is a PBE with principal payoff U^P, so d^* is an optimal direct mechanism in which the expert reports truthfully. □

Lemma 3 allows restriction to truthful mechanisms henceforth. The next lemma says that after a truthful report $r_t = 0$, it is without loss to equalize the continuation values across realized states.

Lemma 4 ($r = 0$ continuations at the optimum)**.** *Let $V \subset \mathbb{R}^2$ be the equilibrium payoff set and let $f : [\underline{V}, \bar{V}] \to \mathbb{R}$ denote the upper Pareto frontier $f(\sigma) = \max\{u : (u, \sigma) \in V\}$, which is concave by convexity of V. Among optimal truthful mechanisms, there exists one for which $\sigma^{0,R} = \sigma^{0,P}$ at every active on-path history.*

Proof. Fix an optimal truthful mechanism M with active-state principal value V^*. At any active history h^t, the principal's stage payoff at the $r_t = 0$ contingency is 0 irrespective of q_t; his continuation contribution is

$$\beta_0 \, f\big(\sigma^{0,R}(h^t)\big) + \beta_1 \, f\big(\sigma^{0,P}(h^t)\big)$$

where $(\beta_0, \beta_1) = \big(p(1-q)/(1-\kappa), (1-p)q/(1-\kappa)\big)$. By concavity of f, this is at most $f(\bar{\sigma})$ with $\bar{\sigma} \equiv \beta_0 \sigma^{0,R}(h^t) + \beta_1 \sigma^{0,P}(h^t)$, with equality iff $\sigma^{0,R}(h^t) = \sigma^{0,P}(h^t)$ or f is affine on the relevant interval.

Hence the modification that replaces both continuations after $(r_t = 0, q_t)$ by a single continuation with value $(f(\bar{\sigma}), \bar{\sigma}) \in V$:

(i) Weakly increases the principal's payoff at h^t.

(ii) Preserves (TT0): truthful continuation at $\eta_t = 0$ is $\bar{\sigma}$ before and after.

(iii) Shifts the (TT1) RHS by $\Delta = \delta q(1-q)(2p-1)[\sigma^{0,R} - \sigma^{0,P}]/[\kappa(1-\kappa)]$.

If $\Delta \leq 0$ the modification preserves (TT1) and is feasible. If $\Delta > 0$, the original M has $\sigma^{0,R} > \sigma^{0,P}$. Because the principal's continuation at the $r = 0$ node is mechanism-irrelevant on stage, M cannot strictly improve on the modified mechanism via this asymmetry; any payoff M achieves through asymmetric continuations is therefore attainable in the limit of a sequence of feasible modifications shrinking $\sigma^{0,R} - \sigma^{0,P}$ in steps of size $\alpha_n > 0$ with $\sum_n \alpha_n \Delta < s$, where $s > 0$ is the slack of (TT1) at M (which is strictly positive by Lemma 2). Compactness of V delivers an optimum with $\sigma^{0,R} = \sigma^{0,P}$. □

Lemma 5 (Binding constraint at $\eta = 0$)**.** *At any optimal truthful mechanism satisfying Lemma 4, the truth-telling constraint at $\eta_t = 0$ binds at every on-path history where the principal has not permanently ceased funding.*

Proof. Under Assumption 1, the principal's stage payoff from truthful play is strictly maximized at $(\rho^0, \rho^1) = (0, 1)$: funding when $r_t = 1$ yields expected stage payoff $(p + q - 1)/\kappa > 0$, while funding when $r_t = 0$ yields expected stage payoff $-(p-q)/(1-\kappa) < 0$. The only on-path randomization in the mechanism is therefore the trigger probability after $(r_t = 1, q_t = 0)$. By Lemma 4 we may take $\sigma^{0,R} = \sigma^{0,P} = \bar{\sigma}$.

The principal's continuation value from any active history h^t at which the trigger probability is ε takes the form

$$U^P(\varepsilon) = \frac{(1-\delta)(p+q-1)}{(1-\delta) + \delta\varepsilon(1-p)(1-q)},$$

which is strictly decreasing in ε:

$$\frac{dU^P}{d\varepsilon} = -\frac{(1-\delta)(p+q-1)\,\delta(1-p)(1-q)}{\big[(1-\delta)+\delta\varepsilon(1-p)(1-q)\big]^2} < 0.$$

The (TT0) slack at h^t is continuous and strictly increasing in ε (since increasing the trigger probability tightens the expert's deviation continuation more than her truthful continuation; this is the content of (2)).

Suppose for contradiction that (TT0) is slack at some active on-path history h^t. Then $\varepsilon(h^t) > \varepsilon(\lambda)$, since (TT0) binds at $\varepsilon = \varepsilon(\lambda)$ and is monotone in ε. Reduce $\varepsilon(h^t)$ by a small $\Delta > 0$: by continuity (TT0) remains satisfied, and U^P strictly increases at h^t, contradicting optimality of the mechanism.

Hence (TT0) binds at every active on-path history. □

Proof of Theorem 2. The proof works recursively via the self-generation technique of Abreu et al. (1990). Let $\mathcal{V} \subset \mathbb{R}^2$ denote the set of pairs $(U^P, U^A) \in \mathbb{R}^2$ of principal and expert equilibrium payoffs achievable by some truthful PBE of the repeated game, with the expert's value computed at the start of any period before the signal is drawn. The set $\mathcal{V}$ is the largest bounded self-generating set, and it is compact and convex (by standard arguments; convexity uses the public randomization device x_t).

By Lemmas 3–5, an optimal mechanism can be decomposed recursively: at every active on-path history, the principal chooses

(a) a period-t funding rule $(\rho^0, \rho^1) \in [0,1]^2$, and

(b) continuation payoffs $(\bar{v}, \sigma_R^1, \sigma_P^1) \in \mathbb{R}^3$ such that $(\cdot, \bar{v})$, $(\cdot, \sigma_R^1)$, $(\cdot, \sigma_P^1)$ each project into $\mathcal{V}$ as achievable expert-continuation values,

to maximize the principal's stage-plus-continuation payoff subject to the binding truth-telling constraint at $\eta_t = 0$:

$$\delta\bar{v} \;=\; (1-\delta)\,\frac{(1-p)q + (1-2\lambda)\,p(1-q)}{1-\kappa} \;+\; \delta\left[\frac{(1-p)q}{1-\kappa}\,\sigma_R^1 \;+\; \frac{p(1-q)}{1-\kappa}\,\sigma_P^1\right], \tag{6}$$

with the convention that the binding constraint is rewritten in (3)'s form after substitution.

Step 1: Optimal stage funding. The principal's stage payoff from truthful play is

$$\mathbb{P}(\eta_t = 1)\cdot\rho^1\cdot\big[\mathbb{P}(q_t = 1 \mid \eta_t = 1) - \mathbb{P}(q_t = 0 \mid \eta_t = 1)\big] + \mathbb{P}(\eta_t = 0)\cdot(-\rho^0)\cdot\big[\mathbb{P}(q_t = 0 \mid \eta_t = 0) - \mathbb{P}(q_t = 1 \mid \eta_t = 0)\big].$$

Under Assumption 1, the first bracket equals $(p+q-1)/\kappa > 0$ and the second equals $(p-q)/(1-\kappa)$, which has the sign of $p-q$. The principal strictly prefers $\rho^1 = 1$ and $\rho^0 = 0$ on the stage, modulo incentive effects on (6).

Step 2: Bang-bang structure of continuations. Let $\overline{V}$ and $\underline{V}$ denote, respectively, the maximum and minimum of $\{u : (\cdot, u) \in \mathcal{V}\}$: the best and worst expert continuation values achievable. Under

truthful play, $\underline{V} = 0$, since the principal can always commit to never fund again (yielding zero for both players), and $\overline{V} > 0$ equals the on-path truthful value of AUNT itself (or any equally-good mechanism).

The principal wishes to minimize the probability of triggering the low-payoff continuation subject to (6). Fix $\bar{v}$ and define the "punishment budget" as the expected reduction in the expert's continuation on the event $\{r_t = 1\}$:

$$B \;\equiv\; \bar{v} - \left[\kappa^{-1} pq\, \sigma_R^1 + \kappa^{-1}(1-p)(1-q)\, \sigma_P^1\right] \;=\; \bar{v} - \frac{pq\, \sigma_R^1 + (1-p)(1-q)\, \sigma_P^1}{\kappa}.$$

The constraint (6) pins down a required value of B; given the linearity of constraint and objective in (σ_R^1, σ_P^1), the principal's optimization over continuation payoffs is a linear program on the compact convex set $\mathcal{V}$. The solution lies at an extreme point: either $\sigma_P^1 = \underline{V} = 0$ with σ_R^1 interior, or both at corners of $\mathcal{V}$.

Consider $\sigma_P^1 = 0$. The required punishment budget B determines σ_R^1 implicitly. The principal's continuation payoff is maximized by setting σ_R^1 as large as possible (since higher σ_R^1 corresponds to continuation mechanisms with higher principal value under truthful play), which means $\sigma_R^1 = \bar{v}$. In that case, the event $\{r_t = 1, q_t = 0\}$ triggers the zero-continuation with probability

$$\varepsilon \;\equiv\; \frac{\bar{v} - \sigma_P^1}{\bar{v}} \;=\; \mathbb{P}(\text{trigger} \mid r_t = 1,\, q_t = 0),$$

which is exactly the AUNT structure: trigger with probability ε after a wrong $r = 1$ recommendation.

Step 3: Optimal $\bar{v}$ and ε. With $\sigma_P^1 = 0$ and $\sigma_R^1 = \bar{v}$ (on the "on-path active" branch), constraint (6) becomes exactly (2), and $\bar{v} = \sigma_R^*$ satisfies the promise-keeping equation (1). Choosing ε to make (2) bind yields $\varepsilon = \varepsilon(\lambda)$ from (3). By Lemma 1, $\varepsilon(\lambda) \leq 1$ under the patience condition (4), so the mechanism is feasible.

Step 4: No alternative extreme dominates. The only alternative extreme points of $\mathcal{V}$ for (σ_R^1, σ_P^1) under truthful play are: (i) $\sigma_P^1 = \overline{V}, \sigma_R^1 = \overline{V}$ (no punishment: infeasible given binding (2)); (ii) $\sigma_R^1 = 0, \sigma_P^1 = 0$ (permanent exclusion regardless of q_t: dominated since σ_R^1 does not appear in the constraint in a way that would require it to be low); (iii) $\sigma_R^1 < \overline{V}, \sigma_P^1 = 0$: dominated by AUNT because reducing σ_R^1 below $\overline{V}$ tightens (6) without payoff-relevant benefit, forcing a larger ε and strictly reducing the principal's continuation value. Hence AUNT is the unique optimum up to payoff-equivalent relabelings.

Applied recursively (at every active on-path history, the continuation mechanism's "active" branch is itself AUNT), this yields the stationary AUNT mechanism as the unique optimal mechanism. □

5.4 Properties of AUNT

The optimal mechanism has several features worth noting.

Remark 4 (Markov structure)**.** AUNT is stationary: conditional on being in the active regime, the rule is identical each period. In particular, history dependence is binary (active versus absorbing). This stands in contrast to mechanisms with finite K-period bans, which require finer history-dependence and, as Theorem 2 shows, are strictly dominated.

Remark 5 (Observability)**.** AUNT uses only realized qualities of *funded* projects. This is reassuring: in many applications (grant review, medical trials, investment decisions) the counterfactual quality of rejected proposals is not observable. A more general mechanism that conditioned on the quality of rejected projects cannot strictly improve on AUNT, by Theorem 2.

Proposition 1 (Opposite bias)**.** *Suppose the expert's period payoff is* $u(d_t, q_t) = \lambda u^P(d_t, q_t) + (1-\lambda)\mathbf{1}\{d_t = 0\}$*, so the expert is biased toward rejecting projects rather than funding them. Assume Assumption 1 and* $\lambda < \bar{\lambda}^-(p,q) \equiv \frac{\kappa}{2pq}$*, and define the patience threshold* $\delta^{**}(p,q) \equiv \frac{\kappa}{pq+(1-q)\kappa}$*. For* $\delta \geq \delta^{**}(p,q)$*, the following* mirror *mechanism* $d^{AUNT^-}(\varepsilon^-(\lambda))$ *is optimal: starting in the active state, the principal implements* $d_t = r_t$ *each period; after any period with* $r_t = 0$ *and* $q_t = 1$ *(a rejection recommendation whose project turned out to be good), the principal transitions with probability* $\varepsilon^-(\lambda)$ *to an absorbing state in which all future projects are funded regardless of recommendation. The optimal punishment probability is*

$$\varepsilon^-(\lambda) = \frac{(1-\delta)\big[(1-p)(1-q) + (1-2\lambda)pq\big]}{\delta\, q(1-q)(2p-1)},$$

and the principal's payoff is

$$U^P\big(d^{\text{AUNT}^-}(\varepsilon^-(\lambda))\big) = \frac{(1-\delta)(p+q-1) + \delta(1-p)q\,\varepsilon^-(\lambda)(2q-1)}{(1-\delta) + \delta(1-p)q\,\varepsilon^-(\lambda)}.$$

Under truthful play, the trigger event $(r_t = 0, q_t = 1)$ *has unconditional probability* $(1-p)q$ *per active period.*

Remark 6. The constant $(2q-1)$ in the numerator of U^P reflects the principal's flow payoff in the absorbing "always fund" state. Setting $q > 1/2$ makes this term positive, so the absorbing state is less punishing for the principal than under the original AUNT; the trigger probability $\varepsilon^-(\lambda)$ compensates accordingly. As $\delta \to 1$, $U^P \to 2q-1$ (the absorbing flow), not 0.

Proof of Proposition 1. Define the relabeling $\tilde{r}_t \equiv 1 - r_t$, $\tilde{d}_t \equiv 1 - d_t$, $\tilde{q}_t \equiv 1 - q_t$, $\tilde{\eta}_t \equiv 1 - \eta_t$. Signal precision is preserved, $\mathbb{P}(\tilde{\eta}_t = \tilde{q}_t \mid \tilde{q}_t) = p$; the relabeled prior is $\tilde{q} \equiv 1 - q$ and p is unchanged.

The expert's payoff decomposes as

$$u(d_t, q_t) = \lambda(1-\tilde{d}_t)(1-2\tilde{q}_t) + (1-\lambda)\tilde{d}_t = \underbrace{\lambda\tilde{d}_t(2\tilde{q}_t - 1) + (1-\lambda)\tilde{d}_t}_{\text{AUNT expert payoff}} + \lambda(1-2\tilde{q}_t),$$

where the residual $\lambda(1-2\tilde{q}_t)$ is independent of $\tilde{d}_t$ and does not affect best-replies. The principal's

payoff decomposes as

$$u^P(d_t, q_t) = (1-\tilde{d}_t)(1-2\tilde{q}_t) = \underbrace{\tilde{d}_t(2\tilde{q}_t - 1)}_{\text{AUNT principal payoff}} + (1-2\tilde{q}_t),$$

contributing an additive constant $\mathbb{E}[1-2\tilde{q}_t] = 2q-1$ to the discounted-average principal payoff.

Apply Theorem 2 to the relabeled problem at $(p, 1-q, \lambda)$, noting that $\kappa(p, 1-q) = 1-\kappa$. The optimal trigger probability is

$$\varepsilon^-(\lambda) = \varepsilon(\lambda; p, 1-q) = \frac{(1-\delta)\big[(1-p)(1-q) + (1-2\lambda)pq\big]}{\delta\, q(1-q)(2p-1)},$$

the feasibility threshold is

$$\delta^{**}(p,q) = \delta^*(p, 1-q) = \frac{\kappa}{pq + (1-q)\kappa},$$

and the relabeled AUNT value is $(1-\delta)(p-q)/[(1-\delta) + \delta(1-p)q\,\varepsilon^-(\lambda)]$. Adding the constant $(2q-1)$ gives

$$U^P = (2q-1) + \frac{(1-\delta)(p-q)}{(1-\delta) + \delta(1-p)q\,\varepsilon^-(\lambda)} = \frac{(1-\delta)(p+q-1) + \delta(1-p)q\,\varepsilon^-(\lambda)(2q-1)}{(1-\delta) + \delta(1-p)q\,\varepsilon^-(\lambda)}.$$

The trigger event $(r_t = 0, q_t = 1)$ has unconditional probability $(1-p)q$ under truthful play, hence the $(1-p)q$ (not pq) in the denominator. □

Remark 7 (Comparative statics)**.** The punishment probability $\varepsilon(\lambda)$ is decreasing in λ (more aligned experts need less discipline), decreasing in p (more precise signals require less punishment per unit of deterrence), decreasing in δ (patient experts respond more to future threats), and increasing in $(1-q)$ when the relevant region is interior (lower-quality priors require stronger discipline, since misreporting is more tempting stage-by-stage). The principal's value U^P is strictly increasing in λ, p, and δ.

6 Conclusion

This paper studies how a principal can discipline a known-biased expert in a repeated advice setting when monetary transfers are unavailable. The optimal mechanism is simple, stationary, and Markovian: rubber-stamp recommendations on-path, and with a carefully calibrated probability commit to permanent silence after any funded project that turns out to be bad. The punishment is extreme but rare: along the equilibrium path, it occurs only after genuine but unlucky recommendations, and its probability is tuned exactly to make the expert's binding incentive constraint just bind.

The paper complements folk-theorem results in dynamic cheap talk and repeated Bayesian

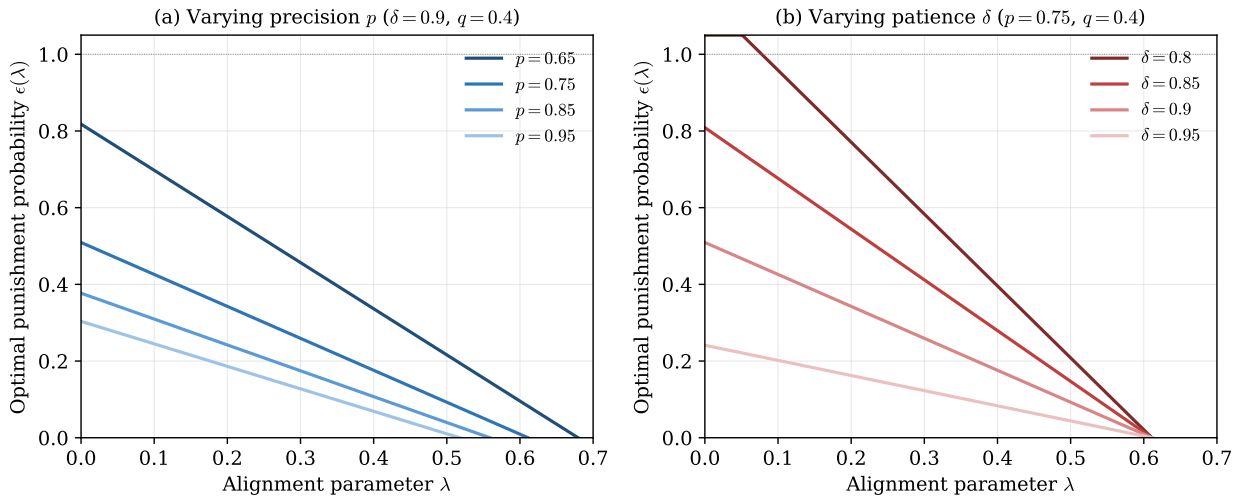

Figure 1: Optimal punishment probability $\varepsilon(\lambda)$ as a function of the alignment parameter λ. Panel (a) fixes $\delta = 0.9$ and $q = 0.4$ and varies the signal precision p: higher precision uniformly lowers $\varepsilon(\lambda)$. Panel (b) fixes $p = 0.75$ and $q = 0.4$ and varies the discount factor δ: patience reduces the punishment needed to sustain truth-telling, and for δ below the threshold $\delta^*(p, q)$ in (4) the line for $\varepsilon(0)$ exceeds 1, violating feasibility (visible for $\delta = 0.8$). Both panels display ε on the region $\lambda \in [0, \bar{\lambda}(p, q)]$; for $\lambda \geq \bar{\lambda}(p, q)$, rubber-stamping attains the first-best and $\varepsilon(\lambda) = 0$.

games (e.g., Margaria and Smolin, 2018) by obtaining an exact optimal-mechanism characterization in a repeated communication environment with principal commitment and observable outcomes. It offers a commitment-based complement to reputation-based approaches for disciplining biased information providers over time. And it extends the review-strategy tradition from Green and Porter (1984); Radner (1985); Abreu et al. (1990) to an information-elicitation rather than effort-provision environment.

Several extensions of the model are natural and are pursued in companion work. First, when the bias λ is privately known to the expert rather than observable to the principal, the screening problem turns out to be structurally constrained: under common signal precision, no incentive-compatible mechanism can strictly improve on the type-blind AUNT calibrated to the worst-case bias. Second, when multiple reviewers are available, principal commitment enables cross-reviewer discipline as a substitute for intertemporal punishment, strictly improving on single-reviewer AUNT even when reviewers share the same bias direction.

A Almost-Perfect Precision

The next result establishes continuity of the first-best at $p = 1$: when signal precision is near-perfect and the expert is patient enough, the principal can approximate the first-best arbitrarily closely using the UNIT mechanism.

Theorem 3 (Near-first-best under almost-perfect precision)**.** *Fix any $\eta > 0$. If $\delta > 1/(1+q)$, then there exists $\gamma \in (0,1)$ such that for any $p \geq 1-\gamma$, the mechanism d^{UNIT} yields a principal payoff satisfying*

$$U^P(d^{UNIT}) \geq U^{FB} - \eta.$$

Proof. At $p = 1$, $\kappa = q$ and $\bar{\lambda}(p,q) = 1/2$. Under UNIT with $p = 1$, the expert's signal is perfectly informative; after $\eta_t = 0$, reporting $r_t = 0$ yields continuation σ_R^* (with promise-keeping (1) evaluated at $\varepsilon = 1$, $p = 1$), while reporting $r_t = 1$ yields stage utility $(1-2\lambda)$ and continuation 0 (the misreport reveals the bad project deterministically). Truth-telling at $\eta_t = 0$ therefore requires

$$\delta\, \sigma_R^* \geq (1-\delta)(1-2\lambda),$$

which rearranges to

$$\delta \geq \frac{1-2\lambda}{(1-2\lambda) + q(1-2\lambda+2\lambda q)} = \frac{1}{1+q} \quad \text{when } \lambda = 0.$$

For any $\lambda \geq 0$, $\delta \geq 1/(1+q)$ suffices.

By continuity of all payoff expressions in p, for any $\eta > 0$ there exists $\gamma > 0$ such that for $p \geq 1-\gamma$, the incentive constraint still holds and

$$U^P(d^{\text{UNIT}}) = \frac{(1-\delta)(p+q-1)}{(1-\delta)+\delta(1-p)(1-q)} \xrightarrow{p\to 1} p+q-1 = U^{FB},$$

with the gap bounded by η for small enough γ. □

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